Impulse Responses and Variance Decomposition Summary — Part 3

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At a Glance

Part 3 turns tangled shocks into independent ones (orthogonalization), then reveals the path along which those shocks spread through the system (IRF) and the sources of forecast uncertainty (FEVD). The conclusions come down to three points.

1. **Orthogonalize the shocks first.** VAR residuals are correlated with one another, so one variable cannot be shocked on its own. The Cholesky factorisation creates mutually independent 'structural shocks'. The result, however, depends on the ordering of the variables.
2. **The IRF shows the shape of a shock; FEVD shows its share.** The IRF shows how each variable responds over time to a single shock; FEVD shows what percentage of each variable's forecast error variance comes from which shock.
3. **It is description, not proof.** IRFs and FEVDs from a reduced-form VAR rest on an ordering assumption, so report their sensitivity to alternative orderings and do not read them as causal.

1. The Problem: VAR Shocks Are Entangled

VAR residuals are correlated within the same period, so the experiment "shock variable 1 only" cannot even be defined in the reduced-form model. This entanglement has to be resolved before shock analysis is possible.

The question that follows from Parts 1 and 2

Part 1 built a VAR and forecast jointly. Part 2 tested who forecasts whom. Part 3 asks a deeper question: **when a shock hits one variable, how does its effect spread through the whole system over time?**

The residual covariance is not diagonal

The residual covariance matrix estimated in Part 1 generally has non-zero off-diagonal elements (the 'instantaneous causality' of Part 2).

$$\Sigma = \mathrm{Cov}(\boldsymbol{\varepsilon}_t) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}, \qquad \sigma_{12} \neq 0$$

- In real markets, a shock to one series does not arrive alone. Interest-rate shocks and stock-price shocks often arrive on the same day.
- So if we give ε1 a one-unit shock, we ignore the movement in ε2 that usually comes with it.
- As a result, the "response to an ε1 shock" is in fact a response to a mixture of two shocks, and the interpretation becomes ambiguous.

What we need is a set of **mutually uncorrelated** shocks. Then moving one leaves the rest unchanged, making a clean 'shock experiment' possible. This transformation is called orthogonalization.

2. Orthogonalization and the Cholesky Factorisation

Rewriting correlated residuals as a combination of mutually independent shocks is orthogonalization, and the standard method is the Cholesky factorisation. The price is an assumption about the ordering of the variables.

The idea of orthogonalization

Express the observed residuals ε_t as a mixture of 'structural shocks' u_t that are mutually uncorrelated and have variance 1.

$$\boldsymbol{\varepsilon}_t = P\,\mathbf{u}_t, \qquad \mathrm{Cov}(\mathbf{u}_t) = I \quad \Longrightarrow \quad \Sigma = PP^{\top}$$

- Because the u_t are independent, an experiment that moves only one of them is cleanly defined.
- The matrix P captures how the independent shocks mix into the observed residuals. The equation on the right, Σ = PP^⊤, follows directly from the condition that the covariance of u is the identity matrix.
- Infinitely many P satisfy this condition. We need a rule for choosing one of them.

The Cholesky factorisation: a lower-triangular P

Among the P satisfying Σ = PP^⊤, the Cholesky factorisation picks the **lower-triangular** one. It is the 'triangular square root' of the covariance matrix. Written out for two variables, it is as follows.

$$\begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} u_{1,t} \\ u_{2,t} \end{pmatrix} \quad \Longleftrightarrow \quad \begin{aligned} \varepsilon_{1,t} &= p_{11}\,u_{1,t} \\ \varepsilon_{2,t} &= p_{21}\,u_{1,t} + p_{22}\,u_{2,t} \end{aligned}$$

Its entries come straight from Σ.

$$p_{11} = \sigma_1, \qquad p_{21} = \frac{\sigma_{12}}{\sigma_1}, \qquad p_{22} = \sqrt{\sigma_2^2 - \frac{\sigma_{12}^2}{\sigma_1^2}}$$

For example, if both variables have variance 1 and correlation 0.4, then p11 = 1, p21 = 0.4 and p22 = √(1 − 0.16) ≈ 0.917. One unit of variable 1's structural shock moves variable 1 by 1.0 and variable 2 by 0.4 (the shared part) in the same period.

The ordering problem

A triangular P means assuming a **causal ordering** of the variables. In the equation above, ε1 contains only u1, while ε2 contains both u1 and u2.

- A shock to the first variable can affect all other variables within the same period.
- A shock to a later variable cannot affect an earlier variable within the same period (only from the next period on).
- The same-period movement shared by the two shocks (σ12) **is taken entirely by the variable placed first.**

Changing the ordering reverses this allocation, so the short-run responses and variance shares change (the chart in Section 4). This is not something the data decide; it is an **assumption** the analyst chooses.

There are three ways to deal with it.

1. Set the ordering by economic reasoning. Put the most exogenous variable (one that does not respond immediately to the others) first. Example: policy rate → output → stock returns.
2. Report how much the results change when the ordering is changed (sensitivity). Do not present just one result.
3. For robust analysis, consider alternatives that do not depend on ordering (generalised IRFs, structural identification methods).

3. The Impulse Response Function (IRF)

The impulse response function traces how much a one-unit independent shock to one variable moves every variable over time. It turns the VAR's coefficient matrices into an intuitive story and is the standard format for reporting VAR results.

Starting point: rewriting the VAR as a 'sum of past shocks'

A stable VAR(1) can be written as a weighted sum of past shocks (the moving-average form, MA(∞)). It is the idea from Part 1, "shocks pass on by multiplying by A each period", unrolled all the way.

$$\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} A^h \boldsymbol{\varepsilon}_{t-h} = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \underbrace{A^h P}_{\Theta_h} \mathbf{u}_{t-h}$$

- The second equality substitutes ε = Pu. Each term is now expressed as a weight Θ_h = A^hP on the independent shocks u.
- The stability condition (eigenvalues inside the unit circle) is needed for A^h to shrink to 0 so that this sum converges.

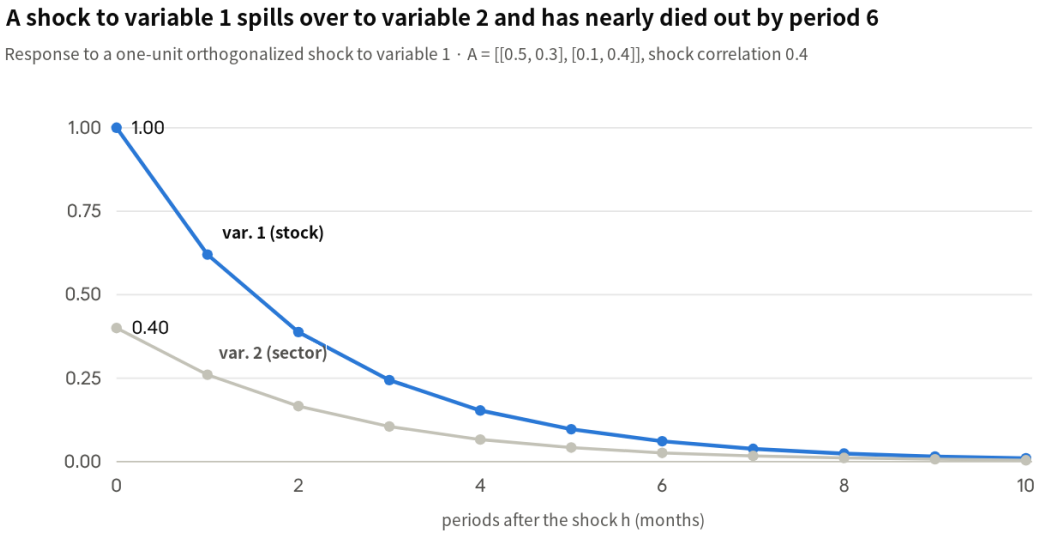
Definition

The IRF is exactly the elements of that weight Θ_h.

$$\mathrm{IRF}_{ij}(h) = \frac{\partial\, y_{i,t+h}}{\partial\, u_{j,t}} = \left[A^h P\right]_{ij} = \left[\Theta_h\right]_{ij}$$

- Give the structural shock u_j of variable j one unit at time t, and read how much variable i has moved h periods later.
- At h = 0, Θ_0 = P, so the same-period response (impact) is simply the columns of the Cholesky matrix P.
- Each extra period multiplies by A once more. The calculation is repeated matrix multiplication.

Example: how a shock spreads and fades



Computed with the two-variable VAR(1) example from the course practice materials · h = 0–10

A shock to variable 1 moves variable 1 by 1.00 and variable 2 by 0.40 (the shared part, the first column of P) in the same period. Each subsequent multiplication by A shrinks them, and by around period 6 both responses fall below 0.06. The speed of decay is set by the eigenvalues of A (0.63, 0.27). The 2×2 IRF grid in the lecture lays out pictures like this in four panels: "2 shocks × 2 responding variables".

How to read an IRF

- **Sign**: does the shock push the variable up or down?
- **Size and peak**: is the response largest immediately, or does it peak a few periods later (overshooting)? Cross responses (variable 2 shock → variable 1) often start at 0 and peak a few periods later.
- **Persistence**: how many periods until it fades? The speed of decay shows the persistence of the system. A stable VAR eventually returns to 0, either smoothly or with oscillations.
- **Confidence band**: an IRF is an estimate, so always plot it with a confidence band. Read a response as meaningful only when the band excludes 0. Bands usually widen at longer horizons.

4. Forecast Error Variance Decomposition (FEVD)

FEVD (forecast error variance decomposition) is a 'variance budget' that splits the uncertainty of a variable's h-step forecast (its forecast error variance) into the percentage coming from each shock. If the IRF shows the **shape** of a shock's effect, FEVD shows its **share**.

The forecast error is the sum of shocks still to come

In the moving-average form of Section 3, what an h-step forecast gets wrong is the shocks that have not yet arrived between now and t+h.

$$\mathbf{Y}_{t+h} - \hat{\mathbf{Y}}_{t+h|t} = \sum_{s=0}^{h-1} \Theta_s \, \mathbf{u}_{t+h-s}$$

Because the structural shocks u are mutually independent with variance 1, the forecast error variance splits neatly into a simple sum of contributions by shock. This is why FEVD also needs orthogonalization: with correlated shocks, covariance terms remain and the variance cannot be attributed to any one shock.

Definition

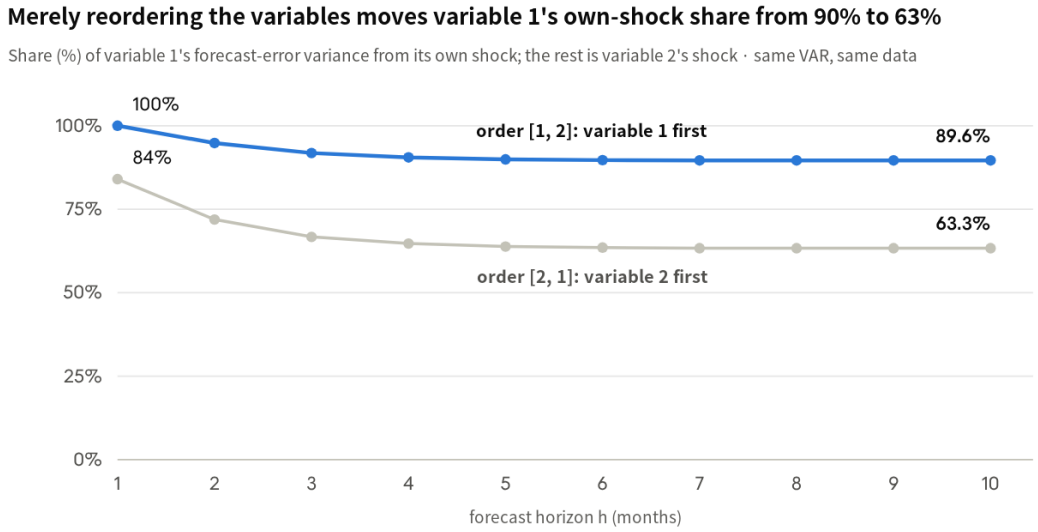
$$\mathrm{FEVD}_{ij}(h) = \frac{\sum_{s=0}^{h-1} [\Theta_s]_{ij}^2}{\sum_{s=0}^{h-1} \sum_{m=1}^{k} [\Theta_s]_{im}^2}, \qquad \sum_{j=1}^{k} \mathrm{FEVD}_{ij}(h) = 1$$

- **Numerator**: the part of variable i's forecast error variance that comes from shock j alone. It is the squared IRF values summed up to horizon h.
- **Denominator**: the total forecast error variance of variable i. It is the sum of the contributions of all shocks, equal to the i-th diagonal element of the forecast error covariance ΣA^iΣ(A^i)^⊤ from Part 1.
- Since it is a ratio, read it as a percentage at each horizon h; summed over all shocks it is always 100%.

The typical shape

- At short horizons, own shocks usually account for most of the variance. Under Cholesky, the first variable is 100% own shock at h = 1.
- As the horizon lengthens, other variables' shocks seep in through the cross effects and their share grows.
- In a stable VAR, the shares converge to fixed values beyond a certain horizon.

Example: same model, different ordering



Computed with the two-variable VAR(1) example from the course practice materials · h = 1–10, two variable orderings

With variable 1 first, 100% of variable 1's uncertainty at h = 1 comes from its own shock, and the share of variable 2's shock grows to about 10% by month 6. In the same model with variable 2 first, the shared same-period movement (correlation 0.4) is credited to variable 2's shock, so the own share starts at 84% and falls to 63%. Not a single data point changed. Only the ordering assumption did.

How to use it

- Rank the sources of uncertainty for each variable. Does the variable move on its own, or is it driven by outside shocks?
- Find which variables are genuinely affected by shocks from elsewhere. For example, if "Y2 shocks explain 60% of Y1's variance", Y1's forecast uncertainty comes mainly from Y2.
- Like the IRF, it depends on the ordering, so report its sensitivity as well. FEVD shares split forecast variance; they are not causal contributions.

5. The VAR Through Three Lenses, and the Bridge to Next Week

A single fitted VAR answers three questions: who forecasts whom (Granger), how shocks spread (IRF), and where uncertainty comes from (FEVD).

Tool	Question it answers	Output	Ordering assumption needed
Granger causality (Part 2)	Does X's past help forecast Y?	Yes / no, and direction	No
Impulse response	How does an X shock	Response path	Yes (orthogonalization)

Tool	Question it answers	Output	Ordering assumption needed
function (IRF)	move Y over time?		
Variance decomposition (FEVD)	What % of Y's uncertainty comes from X shocks?	Share by horizon (%)	Yes (orthogonalization)

The big picture of the week

- **What a VAR provides**: joint forecasts that respect cross-dynamics (Part 1), directional forecasting relationships seen through Granger tests (Part 2), and shock transmission and variance budgets seen through IRF and FEVD (Part 3).
- **Honest caveats**: a reduced-form VAR is a description, not a proof of causation. Structural interpretations such as IRF and FEVD depend on the ordering assumption. Keep the system small, stationary and diagnosed.

On to Week 7: from short-run dynamics to long-run equilibrium

A VAR models how series move together **in the short run**. But a standard VAR requires differenced (stationary) series, so the long-run relationships contained in the levels are thrown away.

- Some pairs share a long-run equilibrium in price levels to which they always return. This is cointegration.
- The vector error correction model (VECM) captures how the series return once they deviate from that equilibrium.
- This 'anchor that pulls back' is the statistical engine of pairs trading and the topic of Week 7.

6. Common Misconceptions and Self-Check

Common misconception	Reality
You can just shock one of the original residuals and read the response	The residuals are correlated with one another. Orthogonalize first (Sections 1 and 2)
The IRF is a single truth regardless of ordering	Cholesky results depend on the ordering. Report the sensitivity (Sections 2 and 4)
IRF and FEVD prove causation	They only describe the reduced-form system; they do not prove a mechanism. FEVD shares split forecast variance (Section 4)

Three questions before wrapping up

1. **Why must VAR shocks be orthogonalized before computing an IRF?** Because the residuals are correlated, so a shock that moves only one of them cannot be defined.

2. **Reversing the variable ordering changed the IRF. Why?** Because the Cholesky factor is lower-triangular, so it assigns the shared same-period movement to the variable placed first.
3. **In the FEVD, Y2 shocks explain 60% of Y1's variance. What does this mean?** It means most of Y1's forecast uncertainty comes from Y2's shocks. This number, too, rests on the ordering assumption.